EXPERIMENT - 10

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1. Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction.
If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints.
The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

Solution:

```
BEGIN TRANSACTION;

INSERT INTO students(name, age, class)
VALUES ('Anisha', 16, 8);
DO $$
BEGIN
   RAISE NOTICE 'Anisha inserted successfully';
END;
$$;
SAVEPOINT SP1;

INSERT INTO students(name, age, class)
VALUES ('Neha', 17, 8);
DO $$
BEGIN
   RAISE NOTICE 'Neha inserted successfully';
END;
$$;
SAVEPOINT SP2;

INSERT INTO students(name, age, class) VALUES
('Anisha', 18, 9);

ROLLBACK TO SAVEPOINT SP2;

DO $$
BEGIN
   RAISE NOTICE 'Failed to insert Anisha. Only failed insert rolled back.';
```

```
END;
$$;
INSERT INTO students(name, age, class)
VALUES ('Mayank', 19, 9);
DO $$
BEGIN
   RAISE NOTICE 'Mayank inserted successfully';
END;
$$;

RELEASE SAVEPOINT SP2;
RELEASE SAVEPOINT SP1;

COMMIT;

SELECT * FROM students;
```